

Le Van Duy

AI Engineer

levduyit@gmail.com | 0985391026 | Ho Chi Minh City | linkedin.com/in/leduy-it

AI Engineer with proven experience in delivering end-to-end computer vision solutions, specializing in OCR, object detection, and document understanding. Familiar with large language models (LLMs) and orchestrator agent frameworks, applying them to enhance automation and data processing workflows. Strong background in designing and deploying high-performance AI systems for both cloud and edge environments.

Work Experience

GMO-Z.com RUNSYSTEM – AI Engineer

Aug 2024 – Present, On-site

- Built and deployed a multilingual OCR platform (Vietnamese/Japanese), designing an iterative data pipeline (continuous enrichment loop) for dataset expansion, model training, and evaluation, with production deployment on cloud and mobile using Triton Inference Server and TensorRT/ONNX, **boosting overall accuracy from 94% → 98% and significantly increasing serving throughput compared to the previous system.**
- Delivered end-to-end computer vision pipelines using modern architectures (YOLO, RT-DETR, Segment Anything), solving practical tasks such as CAD/technical drawing understanding, object measurement, license plate recognition, and structured information extraction from documents, **achieving production-grade accuracy levels required for reliable deployment across diverse real-world document and image types.**
- Orchestrated a production document-parsing pipeline (FastAPI, PostgreSQL, Docker, S3) producing full-content structures (Markdown/HTML/JSON) and schema-based extracts, using managed cloud LLM APIs and in-house VLMs for inference, with normalized outputs ready for downstream RAG systems.
- Implemented an agent-driven Text-to-SQL workflow for stock analysis that converts natural-language queries into validated SQL and normalized tabular outputs for downstream RAG and analytics; the agent integrates semantic table/row search, metadata enrichment (company identifiers and key-term tagging), expert few-shot query templates, and LLM-based self-correction/cross-reflection to validate and normalize results.
- Optimized models for both cloud and edge environments through distillation, quantization, efficient preprocessing/postprocessing, and scalable inference pipelines; introduced dynamic resizing and efficient autoregressive decoder loop management that **reduced OCR inference latency by ~40% and significantly cut compute costs.**
- *The effectiveness of these OCR, document-parsing, agent, and high-performance inference systems has been proven through our customers — including high-demand organizations such as Shinhan Bank, LPBank, ABBANK, Maybank, VCB, MCredit, CITEK, the Ho Chi Minh City Department of Science & Technology, Wifeed, RKKCS (JP), YAMAZEN (JP), and SRA (JP).*

SmartPay JSC – AI Engineer

Mar 2024 – Aug 2024, On-site

- Developed an end-to-end real-time fraud detection pipeline with automated feature engineering and re-training via Apache Airflow, **reducing manual maintenance and ensuring consistent model performance.**
- Shipped an LLM-driven workflow that automatically interprets merchant contracts and configures the company's internal fee-setting system, reducing manual effort for the business operations team.

Technical Skills

- Programming: Python, C++, JavaScript
- AI/ML Frameworks: PyTorch; HuggingFace (Transformers, Datasets, Tokenizers); Sentence Transformers; PaddlePaddle/PaddleOCR; TensorFlow; Scikit-Learn
- LLM & Agent Frameworks: LangChain, LlamaIndex,
- Data & Computation: Pandas, NumPy, CuPy, PostgreSQL, SQLite, Redis, ClickHouse

- Deployment & Orchestration: Docker, FastAPI, Triton Inference Server, Airflow, vLLM
- Model Optimization: ONNX, TensorRT, Knowledge Distillation, Quantization Aware Training, Pruning
- Expertise Domains:
 - Computer Vision (OCR, Document Understanding, Object Detection, Segmentation, Image Processing)
 - OCR Specializations: Multilingual OCR (VN/JP), Text Recognition, Handwriting, MRZ, Layout Analysis
 - LLM Applications: Generative AI, RAG, Text-to-SQL, Structured Extraction, Prompt Engineering
 - Edge & Mobile AI: PaddleLite
- Soft Skills: Problem Solving, End-to-End Project Ownership, Collaboration, Adaptability

Awards

- **3rd Prize** – Naver Vietnamese Handwritten Recognition Track, SoICT Hackathon 2023.
- Core contributor to a project recognized as one of the **Outstanding Innovation & Startup Projects 2025** by the Ho Chi Minh City Department of Science & Technology.

Education

University of Information Technology, VNU-HCM

2020 – 2024

- Bachelor of Science in Computer Science.

Certificates

AI VIET NAM – AI and Data Science Program

2022 – 2023